

protocol: 50% basal dose of conventional urea and 500 ml nano urea through sprays. It found that nano urea significantly reduced the grain yield of rice and wheat by 13% and 17.2%, respectively, compared to 100% conventional urea. There was also 35% and 24% reduction in grain protein content (nutritional value) in rice and wheat, respectively, and a negative impact on chlorophyll content (SPAD), dry matter, tiller density, root growth, and macronutrient content.

In short, Rajeev Sikka, who led the study, said: "If the plant's needs are not fulfilled by the application of 50% dose of nano urea sprays, the yield and quality of the produce will reduce."

This study confirms what farmers have been saying for some years. Balbir Singh Rajewal, founder of the Bharatiya Kisan Union (BKU), says: "Farmers have seen the results, which are not good. So, we are discouraging farmers from buying nano fertilisers."

Sangrur farmer Ghuman points out that the use of nano urea brings down wheat yield from 20-25 quintal per acre by five-six quintal. It is also expensive to hire labour to spray it, unlike granular urea, which a farmer can spray on his own. Besides, one acre needs five bottles of nano spray, not one as is being said, and the sprays get washed away in rain and dew. According to Charuni, it would be wrong to assume that yield improves. "They are misleading us," he says.

Nandula Raghuram, Chair Emeritus of the International



Conflicting results across different trial conditions are not an anomaly in the science of agricultural innovation. That's why continued multi-location, multi-crop, multi-season trials are essential "

KJ PATEL

MANAGING DIRECTOR, IFFCO



Conventional urea works best when applied to the soil, as crops evolved their root system to derive nutrition from it. Maximising availability of urea in soil surrounding the roots has been the focus "

NANDULA RAGHURAM

CHAIR EMERITUS,
INTERNATIONAL NITROGEN
INITIATIVE

Nitrogen Initiative, explains: "Conventional urea works best when applied to the soil, as crops evolved their root system to derive nutrition from it. Therefore, maximising availability of urea in soil surrounding the roots has been the focus of all efforts to improve the efficiency of fertilisers. Foliar spray is at best supplemental, whether in conventional urea or nano urea."

A nano-tech expert working in the fertiliser sector explains on condition of anonymity that plants absorb nutrients from soil through roots, which is the systemic route. "It is like injecting a drug into the bloodstream. Foliar spray has little use. Nanotech may help in delivering micro-nutrients (potassium, phosphorus, etc.) but not nitrogen, which is required in bulk," he says.

Concerns around the utility of nano fertilisers have been flagged in global studies as well. A 2023 publication by the University of Copenhagen concluded that nano urea had "no scientifically proven effects" and was "marketed with misleading and wrong statements about its efficiency, underlying plant uptake pathways, and environmental friendliness."

It also warned: "The expectations raised by IFFCO are far from reality and may lead to large-scale yield losses with serious consequences for food security and livelihood of farmers." It advised "much more priority should be given to scientifically prove their efficacy and mode of action."

For now, the jury is out on the efficiency of nano fertilisers, and only more research and studies would be able to address the gaps highlighted by stakeholders. **BT**